

Mihir Thakur

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SUMMARY

AI Data Scientist specializing in ML solutions, Generative AI systems and cloud infrastructure that has delivered millions in measurable impact.

TECHNICAL SKILLS

Languages and Databases: Python, C/C++, JavaScript, MySQL, MS SQL, PostgreSQL, Supabase
Cloud: Databricks, AWS (S3, EC2, Sagemaker), Azure (AzureML, Data Factory), GCP (VertexAI), Docker
Development: Pandas, scikit-learn, PyTorch, ONNX, PySpark, LangChain, HuggingFace, SHAP, NLTK, FastAPI
Others: Git, GitHub, Linux, Apache Spark, Jupyter, Parquet, MLOps, Copilot, Claude, Jira
Certifications: OCI Generative AI Professional Certificate; Microsoft Certified: Azure Fundamentals (ID: I277-4775)

PROFESSIONAL EXPERIENCE

AI/ML Developer May 2026 - Present
DriverAI Tempe, AZ

- Contributed to image classification with decoupled architecture using **PyTorch** with EfficientNet and YOLO on **AWS SageMaker** and S3.
- Improved synthetic **data augmentation** pipeline for foreground extraction and image transformations.

AI Engineer/Data Scientist June 2025 - December 2025
Ascend Remote, US

- Implemented 10 tools for conversational AI Agentic system using **LangChain** and PostgreSQL.
- Developed full-stack RAG feature using **OpenAI Vector Store** to deliver 6-second context retrieval for AI Agent.
- Analyzed systems and optimized Temporal workflow through asynchronous processing, reducing latency by **200ms**.
- Created **AI evaluation framework** with Playwright MCP and LangChain for testing agent tool calling by defining custom metrics and refactored AI Agent, improving tool calling accuracy by **25%**.

Data Scientist June 2022 - July 2023
LTIMindtree Pune, India

- Implemented 3 data science solutions using Pandas, **PySpark**, and Scikit-learn on **Azure Databricks**.
- Developed ETL pipelines on **Apache Spark** for processing terabytes of data from Data Lake to Delta tables and trained XGBoost/LGBM models to segment customers for targeted marketing, leading to estimated **\$1M revenue increase**.
- Engineered hybrid RoBERTa and rule-based approach for customer segmentation system on **call-center transcription** data, iteratively improving through SME feedback to align model logic.
- Innovated data-driven approach to classify delinquent loan customers for risk-based loan recovery by building end-to-end ML pipelines, employing **MLflow** for lifecycle management, improving recall by relative 10%.
- Delivered \$1.5M reduction in credit losses by validating and deploying new delinquency prediction model that outperformed existing models in production, reducing false negatives by 7% in controlled A/B test.
- Developed dashboards to monitor **model and data drifts** affecting model performance and trigger retraining.
- Communicated model results with **SHAP** findings and data-driven recommendations to business stakeholders and leadership through presentations and PowerBI dashboards.
- Identified and optimized SQL queries using SparkSQL ensuring data quality and security, reducing **runtime by 20%**.
- Collaborated with cross-functional teams to build **cloud infrastructure**, deploying 15 ML pipelines via Azure DevOps.

EDUCATION

Master of Science, Computer Science May 2025
Arizona State University, Tempe, AZ 3.93 GPA
Relevant Coursework: NLP, Statistical ML, Data Visualization, ML Acceleration

Bachelor of Engineering, Computer Science Engineering May 2022
University of Mumbai, Mumbai, India 3.85 GPA

PROJECTS

Benchmarking and Fairness Estimation of Chest X-rays - Benchmarked **transformer models** like ViT, SwinV2, ConvNeXt and REMEDIS by fine-tuning for tuberculosis detection in chest X-rays, achieving 87% accuracy on OOD data with REMEDIS model. Studied **bias and fairness** for all models on gender, age and other criteria.

Hackathons: Zoom Spark Finalists (Zoom Attendance), Voxel AI CV Finalists (BeastWatch - Animal Tracking)